AI versus Other Technical Progress: Interest-Rate and Wage Dynamics in Three Cases

Research note (draft), companion to "AI-Driven Technical Change"

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Abstract

How do the real interest rate and the real wage respond to AI-driven progress compared with other technical progress? The hypothesis is that AI substitutes for labor and possibly for other capital. I study three cases in the production economy of Kiley (2004), using its CES production/CES accumulation pairing. In every case I compare progress that is *disembodied* (in the production function) with progress that is *embodied* (in the accumulation equation), and I scale every shock to raise long-run output by the same 10 percent. Cases 1 and 2 have one capital stock that is a complement to labor ($\varepsilon < 1$) or a substitute for it ($\varepsilon > 1$), and they have exact closed-form solutions. Case 3 has a regular capital stock that is a complement to labor, plus AI capital that starts small and substitutes for both. It has no closed form and is solved numerically. The main results: (1) With the same output gain, capital-specific progress raises the long-run wage by $1/\varepsilon$ times the output gain, against one-for-one for labor-augmenting progress. With substitutes, the wage gain is smaller and the labor share falls. (2) Embodied and disembodied progress have identical long-run effects but different dynamics. A gradual disembodied shock raises r sooner and higher, while embodied progress raises r later and longer, and its wage gains lag by a decade. A surprise permanent step reverses the ranking for r on impact. (3) In case 3, AI progress raises the long-run wage by less (7.6 vs 10.5 percent) and lowers the labor share. It also raises the saving rate more, and it raises the interest rate more on impact.

1 Common structure

Notation follows Kiley (2000). Let $\rho \equiv (\varepsilon - 1)/\varepsilon$. Each capital stock j accumulates through a CES function whose elasticity is that of the production nest it enters:

$$K_j'^{\rho_j} = \bar{A}_j^{\rho_j}\left[(1-\delta_j)K_j^{\rho_j} + \delta_j (I_j/P_j)^{\rho_j}\right], \tag{1}$$

with the Cobb-Douglas form $K_j' = \bar{A}_j K_j^{1-\delta_j}(I_j/P_j)^{\delta_j}$ when $\rho_j = 0$. Here I_j is spending in output units and P_j is the price of a new efficiency unit. The normalization $\delta_j = x_j^{1-\rho_j}/(x_j^{1-\rho_j} + 1 - d_j)$ and $\bar{A}_j^{\rho_j} = (1+g)^{\rho_j - 1}(x_j^{1-\rho_j} + 1 - d_j)$, with $x_j = g + d_j$, makes (1) agree to first order with $K_j' = (1-d_j)K_j + I_j/P_j$ on the balanced growth path, so $Q_j = P_j$ there. Utility is $\sum_t B^t C_t^{1-1/\psi}/(1-1/\psi)$ with intertemporal elasticity ψ. Hours are fixed. Two kinds of progress affect capital j:

- *disembodied*: a factor Γ_j multiplies K_j in production and raises the productivity of the installed stock;
- *embodied*: P_j falls, which raises the efficiency of new investment only.

The benchmark for "other" progress is labor-augmenting Z. The calibration follows the paper: initial $r = 7.5$ percent, capital share 0.282, $d_K = 0.066$, trend growth $g = 1.5$ percent. AI capital has $d_A = 0.20$ and an initial income share of 3 percent. Each shock follows a logistic S-curve centered on year 10 and announced at $t = 0$, or alternatively arrives as a surprise permanent step. Its size is set so that long-run output rises by 10 percent.

2 Cases 1 and 2: one capital stock (exact)

Production is $Y^\rho = a_K(\Gamma K)^\rho + a_L(ZN)^\rho$, with $\varepsilon = 0.5$ (case 1, complements) or $\varepsilon = 1.5$ (case 2, substitutes). Capital accumulates through (1) with the same ρ.

Proposition 1 (Exact dynamics)**.** *Let $\psi = \varepsilon$. For any paths of Γ, P and Z, anticipated or not, the value function is $V_t = \kappa_t K^\rho/\rho + h_t$. The saving rate $s_t = I_t/Y_t$ depends on time alone, not on the capital stock, and solves the backward recursion*

$$\phi_t = B\kappa_{t+1}\bar{A}^\rho\delta P_t^{-\rho}, \qquad s_t = \frac{\phi_t^\varepsilon}{1+\phi_t^\varepsilon}, \qquad \kappa_t = (1+\phi_t^\varepsilon)^{1/\varepsilon}a_K\Gamma_t^\rho + B\bar{A}^\rho(1-\delta)\kappa_{t+1}.$$

Capital follows a law of motion that is linear in K^ρ: $K_{t+1}^\rho = \bar{A}^\rho[(1-\delta+\delta s_t^\rho a_K\Gamma_t^\rho P_t^{-\rho})K_t^\rho + \delta s_t^\rho P_t^{-\rho}a_L(Z_tN)^\rho]$. Prices are $1 + r_{t+1} = B^{-1}(C_{t+1}/C_t)^{1/\varepsilon}$ and $W = a_LZ(Y/ZN)^{1/\varepsilon}$, and the labor share is $a_L(ZN/Y)^\rho$. Labor-augmenting progress leaves s_t unchanged. With constant Γ/P, $s = \Lambda^\varepsilon(\Gamma/P)^{\varepsilon-1}$, where $\Lambda = B\bar{A}^\rho\delta a_K/(1-B\bar{A}^\rho(1-\delta))$.

Proposition 2 (Long run, same output gain γ_Y)**.** *Disembodied and embodied progress have identical long-run effects when $\ln\Gamma = -\ln P$. Let s_K be the initial capital share. The shock size that raises long-run output by γ_Y is*

$$\left(\frac{\Gamma}{P}\right)^{\varepsilon-1} = \frac{1-(1-s_K)e^{-\rho\gamma_Y}}{s_K}.$$

The real interest rate is unchanged in the long run. For capital-specific progress,

$$\Delta\ln W = \frac{\gamma_Y}{\varepsilon}, \qquad \Delta\ln(\textit{labor share}) = -\rho\,\gamma_Y, \qquad \textit{capital share} = s_K(\Gamma/P)^{\varepsilon-1},$$

whereas labor-augmenting progress gives $\Delta\ln W = \gamma_Y$ and an unchanged labor share. These results hold for any ψ.

Proof. On a balanced growth path, $\Gamma F_M = P(r+d)$ with $M = \Gamma K$, so only Γ/P matters. The capital share is $a_K^\varepsilon(\Gamma/P)^{\varepsilon-1}(r+d)^{1-\varepsilon}$, and $(Y/ZN)^\rho = a_L/(1-\text{capital share})$. The wage and labor-share formulas follow from $W = a_LZ(Y/ZN)^{1/\varepsilon}$. □

Proposition 2 is the long-run answer to the question. Fix the gain in output. If AI-type capital is a substitute for labor ($\varepsilon > 1$), workers get less than the output gain: 6.9 percent against 10.5 percent for labor-augmenting progress, and the labor share falls. If it is a complement, workers get more: 22 percent. This holds for embodied and disembodied progress alike, so in the long run the type of progress does not matter. Only its dynamics do.

Dynamics (Figure 1, Table 1).

1. *Interest rates, S-curve.* Disembodied progress raises r sooner and higher than embodied progress in both cases (case 1: 10.8 vs 9.0 percent at the peak; case 2: 7.9 vs 7.7 percent).

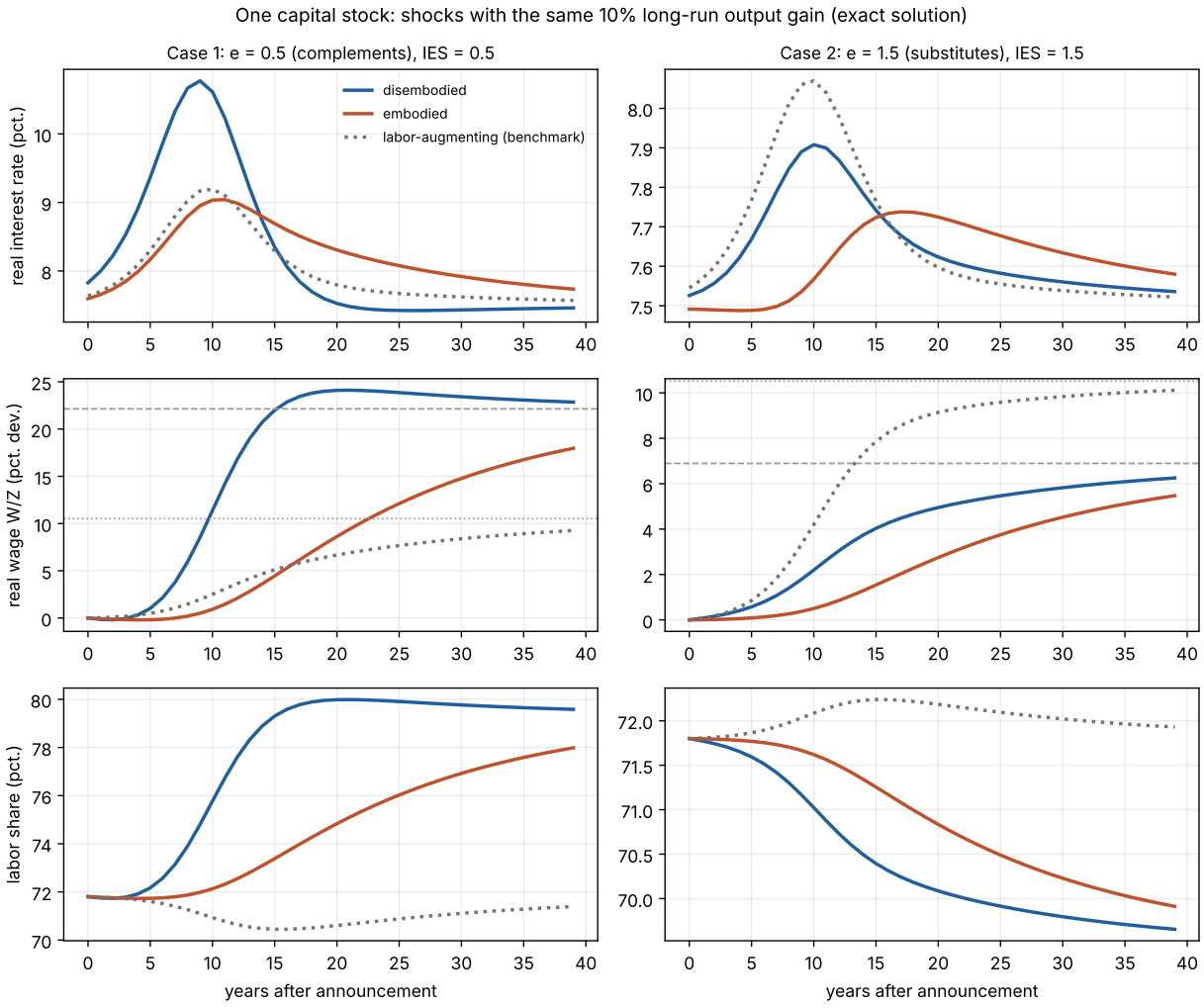


Figure 1: Cases 1 and 2, exact solution ($\psi = \varepsilon$). Shocks have the same 10% long-run output gain. Dashed lines: long-run wage γ_Y/ε; dotted: γ_Y.

Table 1: Cases 1 and 2, exact solution ($\psi = \varepsilon$). Peak r and the wage at t=10 are for the S-curve; r_1 (between years 0 and 1) is for a surprise step. The last column is the cumulative deviation of $\ln(1+r)$ from 7.5% over 150 years.

Case 1, $\varepsilon = 0.5$

	shock (log)	peak r (pct.)	year	r_1, step (pct.)	W, t=10 (pct.)	W, long run (pct.)	labor share long run (pct.)	$\sum \Delta \ln(1+r)$ (pct.)
Disembodied	0.623	10.77	9	6.94	11.4	22.1	79.4	26.5
Embodied	0.623	9.04	11	8.81	0.9	22.1	79.4	29.5
Labor-augmenting	0.100	9.19	10	7.86	2.5	10.5	71.8	20.0

Case 2, $\varepsilon = 1.5$

	shock (log)	peak r (pct.)	year	r_1, step (pct.)	W, t=10 (pct.)	W, long run (pct.)	labor share long run (pct.)	$\sum \Delta \ln(1+r)$ (pct.)
Disembodied	0.160	7.91	10	7.72	2.2	6.9	69.4	6.0
Embodied	0.160	7.74	17	7.90	0.5	6.9	69.4	5.6
Labor-augmenting	0.100	8.07	10	7.61	4.2	10.5	71.8	6.7

The embodied response peaks later (year 11 vs 9 in case 1, year 17 vs 10 in case 2) and lasts longer. In case 2 the interest-rate response to capital-specific progress is *smaller* than to labor-augmenting progress of equal long-run size (peak 8.1 percent). In case 1, disembodied progress raises r more than labor-augmenting progress (peak 9.2 percent). Complementary capital needs a much larger technology shock (0.62 vs 0.16 log points) to deliver the same output gain, and its effect is concentrated in the transition.

2. *Interest rates, surprise step.* The ranking reverses on impact. Embodied progress raises r_1 more (case 1: 8.8 vs 6.9 percent; case 2: 7.9 vs 7.7 percent). With complements, a disembodied step *lowers* r below its initial level. Less investment is needed to reach the new capital stock, so the saving rate falls (from 0.162 to 0.119), consumption jumps and then grows slowly.

3. *Wages.* Embodied progress delays wage gains by about a decade: at year 10 wages are up 0.9 vs 11.4 percent in case 1 and 0.5 vs 2.2 percent in case 2. Disembodied progress with complements pushes the wage temporarily above its long-run gain.

4. *Labor share.* It moves in opposite directions in the two cases, and it moves faster under disembodied progress. Labor-augmenting progress leaves it unchanged in the long run.

Proposition 1 requires $\psi = \varepsilon$, which ties complements to a low intertemporal elasticity and substitutes to a high one. With log utility in both cases (solved numerically, Appendix Table 3), the orderings are unchanged and the magnitudes are similar. In case 1 the interest-rate effects are smaller because the IES is higher.

3 Case 3: regular capital and AI capital (numerical)

Production nests a traditional bundle inside a CES with AI:

$$X = \left[b_K(\Gamma_K K)^{\rho_1} + b_L(ZN)^{\rho_1}\right]^{1/\rho_1}, \qquad Y = \left[c_X X^{\rho_2} + c_A(\Gamma_A A)^{\rho_2}\right]^{1/\rho_2},$$

with $\varepsilon_1 \leq 1$ (regular capital is a complement to labor; Cobb-Douglas in the baseline, $\varepsilon_1 = 0.5$ as an alternative) and $\varepsilon_2 = 1.5$ (AI substitutes for both labor and regular capital). AI starts small, with a 3 percent income share. Each stock accumulates through (1) with its own nest's elasticity. Utility is logarithmic.

Table 2: Case 3. Top: Cobb-Douglas bundle (baseline). Bottom: CES bundle with $\varepsilon_1 = 0.5$. Columns as in Table 1.

Cobb-Douglas bundle

	shock (log)	peak r (pct.)	year	r_1, step (pct.)	W, t=10 (pct.)	W, long run (pct.)	labor share long run (pct.)	$\sum \Delta \ln(1+r)$ (pct.)
K disembodied	0.255	8.34	10	7.71	3.7	10.5	69.6	10.0
K embodied	0.255	7.90	16	8.09	0.7	10.5	69.6	10.0
AI disembodied	1.223	8.16	11	8.29	2.1	7.6	67.8	8.3
AI embodied	1.223	8.06	15	9.18	0.8	7.6	67.8	8.1
Labor-augmenting	0.100	8.34	10	7.71	3.7	10.5	69.6	10.0

CES bundle, $\varepsilon_1 = 0.5$

	shock (log)	peak r (pct.)	year	r_1, step (pct.)	W, t=10 (pct.)	W, long run (pct.)	labor share long run (pct.)	$\sum \Delta \ln(1+r)$ (pct.)
K disembodied	0.623	9.17	9	7.31	12.2	22.1	77.0	13.5
K embodied	0.623	8.30	11	8.17	1.5	22.1	77.0	14.9
AI disembodied	1.327	8.27	11	8.42	2.3	8.5	68.4	8.6
AI embodied	1.327	8.15	15	9.54	0.9	8.5	68.4	8.4
Labor-augmenting	0.100	8.39	10	7.73	3.1	10.5	69.6	10.2

No closed form. The exact solutions require output and continuation value to be linear in each K_j^ρ for a single ρ. Here X^{ρ_2} is not linear in K^{ρ_1} unless $\varepsilon_1 = \varepsilon_2$, which would make AI and regular capital the same kind of input. The one exception is $\varepsilon_1 = \varepsilon_2 = 1$ with log utility, where everything is Cobb-Douglas and AI is neither a substitute nor a complement. Case 3 is therefore solved as a perfect-foresight Newton problem. The residuals are below 10^{-11} in every experiment (Appendix).

A useful equivalence. With a Cobb-Douglas bundle, capital-augmenting progress in K is the same as labor-augmenting progress: $\Gamma_K^\sigma (ZN)^{1-\sigma}$ with σ the bundle's capital share. The two give identical paths for every variable, which the numerical results reproduce exactly. In this baseline, the relevant comparison is *AI progress versus all conventional progress.*

Results (Figure 2, Table 2).

1. *Long-run wages.* With the same 10 percent output gain, AI progress raises the wage by 7.6 percent against 10.5 percent for conventional progress, and the labor share falls from 69.6 to 67.8 percent. With a complementary bundle ($\varepsilon_1 = 0.5$), the contrast is starker. Regular-capital progress raises wages by 22 percent, as in case 1, while AI progress raises them by 8.5 percent.

2. *Size of the technology shock.* Because AI starts small, the same output gain needs a large improvement in AI technology: Γ_A / P_A rises 3.4-fold ($e^{1.22}$), against a 29 percent rise for regular capital. The AI share of income grows along the way.

3. *Interest rates.* For a gradual S-curve, peak effects are similar. Disembodied AI progress peaks slightly below conventional progress (8.2 vs 8.3 percent), and embodied AI progress slightly above embodied regular-capital progress (8.1 vs 7.9 percent). For a surprise step, AI progress raises r *more* on impact: 8.3 vs 7.7 percent if disembodied, and 9.2 vs 8.1 percent if embodied. The AI investment boom is large relative to its initial stock, so the long-run saving rate rises from 0.181 to 0.196, and in the meantime consumption grows faster.

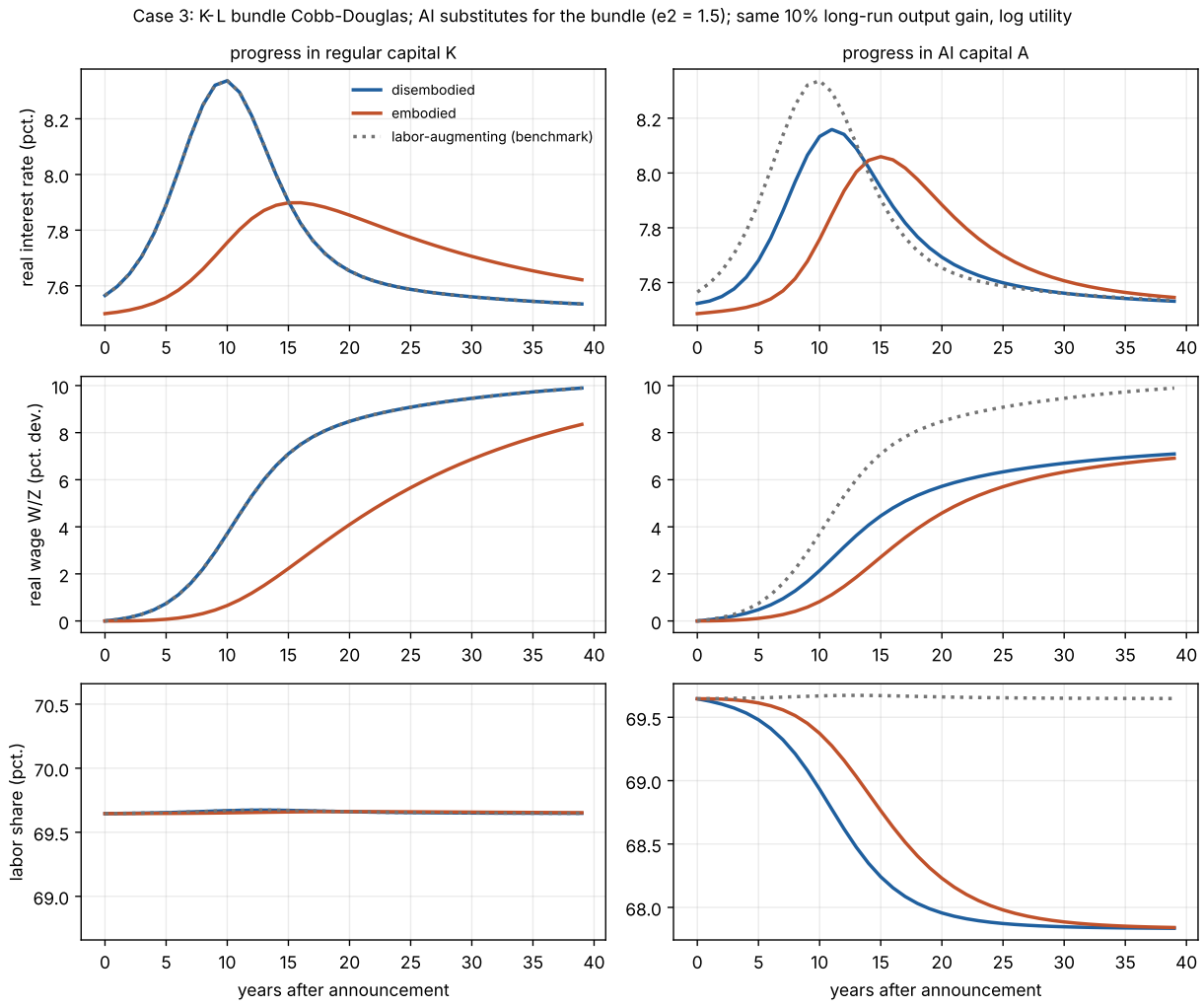


Figure 2: Case 3 (Cobb-Douglas bundle, $\varepsilon_2 = 1.5$, log utility): progress in regular capital versus AI capital, same 10% long-run output gain.

4. *Embodied versus disembodied.* The pattern of cases 1 and 2 carries over. Embodied progress gives a later, more persistent rise in r, a slower rise in wages (0.8 vs 2.1 percent at year 10 for AI) and a larger impact effect after a surprise step.

4 Summary

- *Long run* (closed form in cases 1 and 2). For the same output gain, the wage rises by γ_Y/ε under capital-specific progress, against γ_Y under labor-augmenting progress. If AI substitutes for labor, workers gain less than output and the labor share falls. The long-run interest rate is unchanged by level shifts in technology, and embodied and disembodied progress have identical long-run effects.
- *Dynamics.* Disembodied AI progress (better models applied to installed compute and software) raises interest rates early and lifts wages quickly. Embodied AI progress (cheaper new AI capital) works through an investment boom. Its rise in rates is delayed and more persistent, and its wage gains lag by a decade. A surprise step raises rates more under embodied progress.
- *AI versus conventional progress* (case 3, numerical). AI progress yields smaller wage gains, a falling labor share, a higher saving rate and a larger impact effect on the real rate after news. Its peak interest-rate effect under gradual diffusion is similar to that of conventional progress.

A Robustness and verification

Table 3: Cases 1 and 2 with log utility ($\psi = 1$), solved numerically.

Case 1, $\varepsilon = 0.5$

	shock (log)	peak r (pct.)	year	r_1, step (pct.)	W, t=10 (pct.)	W, long run (pct.)	labor share long run (pct.)	$\sum \Delta \ln(1+r)$ (pct.)
Disembodied	0.623	9.21	9	7.20	12.3	22.1	79.4	13.6
Embodied	0.623	8.32	10	8.20	1.5	22.1	79.4	15.0
Labor-augmenting	0.100	8.41	10	7.69	3.1	10.5	71.8	10.3

Case 2, $\varepsilon = 1.5$

	shock (log)	peak r (pct.)	year	r_1, step (pct.)	W, t=10 (pct.)	W, long run (pct.)	labor share long run (pct.)	$\sum \Delta \ln(1+r)$ (pct.)
Disembodied	0.160	8.07	10	7.80	2.1	6.9	69.4	8.8
Embodied	0.160	7.84	17	8.05	0.4	6.9	69.4	8.4
Labor-augmenting	0.100	8.30	10	7.66	4.0	10.5	71.8	9.8

`cases.py` computes everything. For cases 1 and 2 it compares the backward-recursion saving rates of Proposition 1 with a generic Newton solve of the Euler equations started 5 percent away. It checks the shock sizes and long-run wage formulas of Proposition 2, and it reports Newton residuals for case 3.

References

Kiley, M. T. (2000). "Stock Prices and Fundamentals in a Production Economy." FEDS Working Paper 2000-05, Federal Reserve Board.

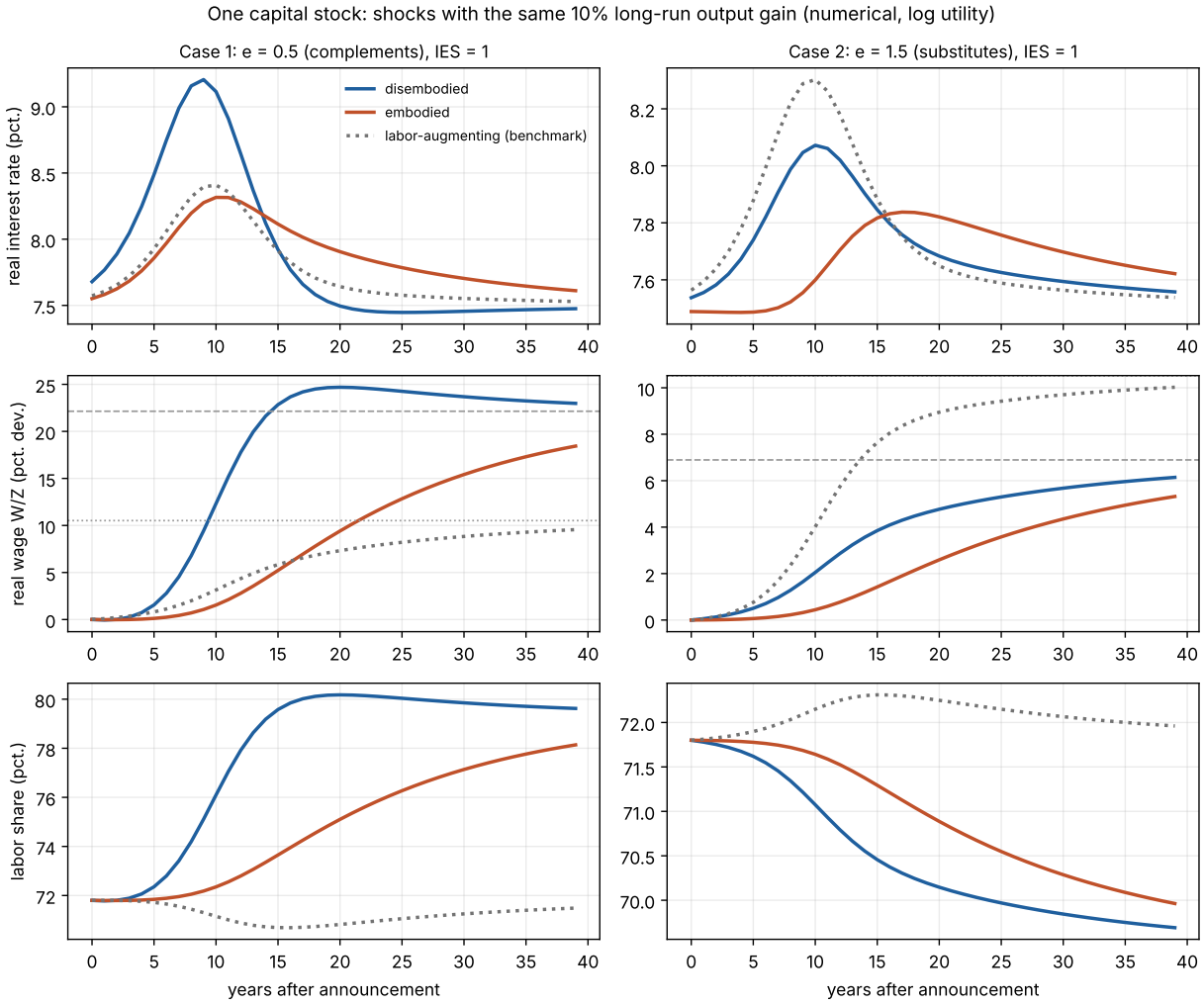


Figure 3: Cases 1 and 2 with log utility ($\psi = 1$).

Kiley, M. T. (2004). "Stock Prices and Fundamentals: A Macroeconomic Perspective." *Journal of Business* 77(4): 909–936.

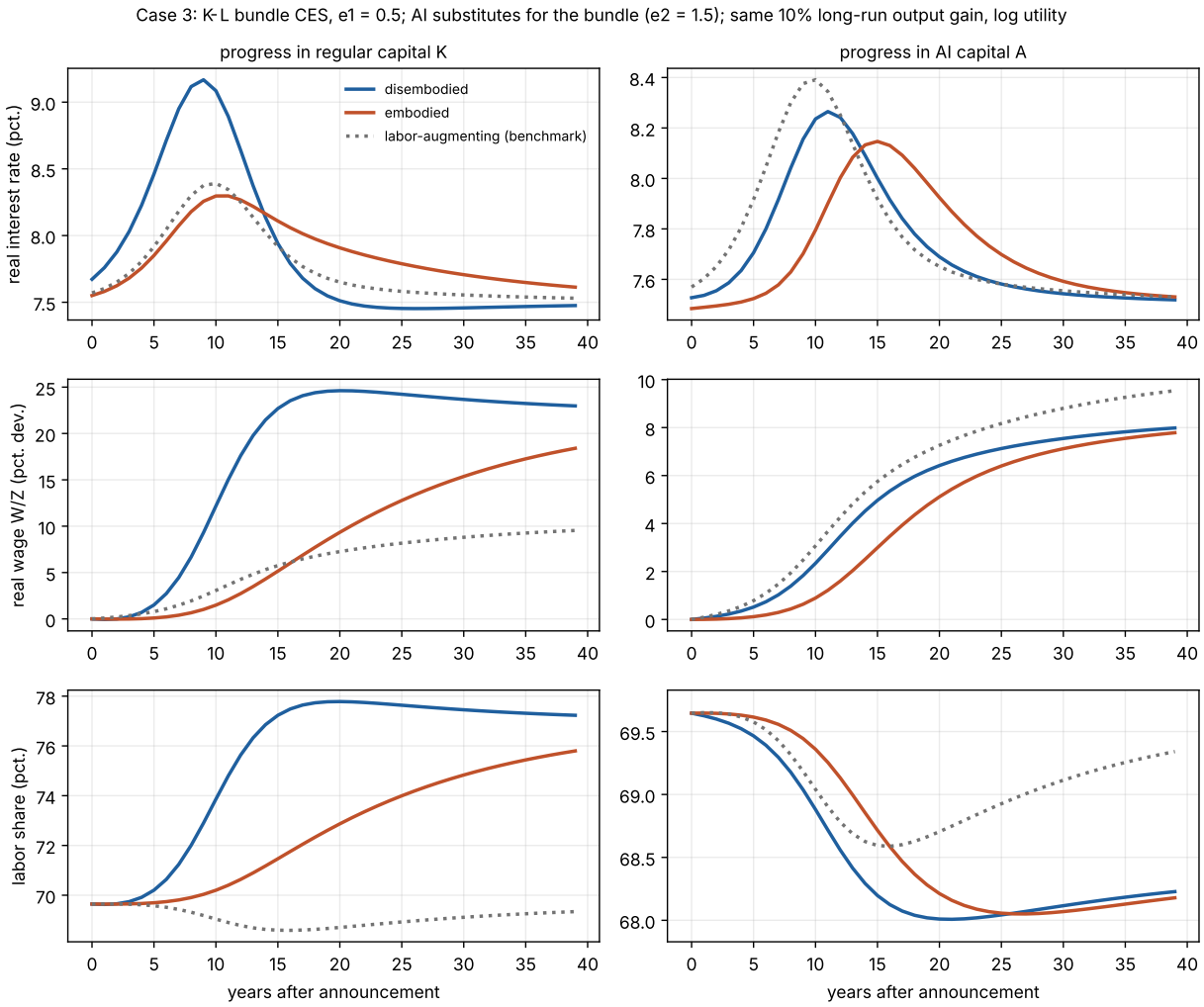


Figure 4: Case 3 with a CES bundle ($\varepsilon_1 = 0.5$).

Table 4: Verification.

Check	max abs. error
case1: shock size = closed form (dis)	6.7e-16
case1 dis S: generic = exact s	8.6e-16
case1 dis step: generic = exact s	8.7e-16
case1: shock size = closed form (emb)	4.4e-16
case1 emb S: generic = exact s	8.7e-16
case1 emb step: generic = exact s	8.7e-16
case1 Z S: generic = exact s	1.1e-15
case1 Z step: generic = exact s	1.1e-15
case1 IES=exact: long-run ln W = target/e (dis)	1.7e-16
case1 IES=exact: long-run ln W = target/e (emb)	1.9e-16
case1 IES=exact: long-run ln W = target (Z)	6.9e-17
case1 IES=log: long-run ln W = target/e (dis)	1.7e-16
case1 IES=log: long-run ln W = target/e (emb)	1.9e-16
case1 IES=log: long-run ln W = target (Z)	6.9e-17
case2: shock size = closed form (dis)	6.7e-16
case2 dis S: generic = exact s	1.2e-12
case2 dis step: generic = exact s	1.2e-12
case2: shock size = closed form (emb)	4.9e-14
case2 emb S: generic = exact s	1.1e-12
case2 emb step: generic = exact s	1.3e-12
case2 Z S: generic = exact s	1.1e-12
case2 Z step: generic = exact s	1.1e-12
case2 IES=exact: long-run ln W = target/e (dis)	1.4e-16
case2 IES=exact: long-run ln W = target/e (emb)	1.2e-15
case2 IES=exact: long-run ln W = target (Z)	6.9e-17
case2 IES=log: long-run ln W = target/e (dis)	1.4e-16
case2 IES=log: long-run ln W = target/e (emb)	1.2e-15
case2 IES=log: long-run ln W = target (Z)	6.9e-17
case3: initial steady state reproduces calibration	1.4e-17
case3 disK S: Newton residual	1.9e-14
case3 disK step: Newton residual	5.9e-12
case3 embK S: Newton residual	1.8e-15
case3 embK step: Newton residual	5.0e-15
case3 disA S: Newton residual	4.8e-13
case3 disA step: Newton residual	2.9e-15
case3 embA S: Newton residual	6.9e-13
case3 embA step: Newton residual	5.9e-14
case3 Z S: Newton residual	1.9e-14
case3 Z step: Newton residual	5.9e-12
case3 ces: initial steady state reproduces calibration	1.4e-17
case3 ces disK S: Newton residual	6.5e-12
case3 ces disK step: Newton residual	2.8e-15
case3 ces embK S: Newton residual	3.4e-15
case3 ces embK step: Newton residual	2.6e-13
case3 ces disA S: Newton residual	6.5e-13
case3 ces disA step: Newton residual	3.0e-13
case3 ces embA S: Newton residual	8.1e-13
case3 ces embA step: Newton residual	5.8e-12
case3 ces Z S: Newton residual	4.2e-13
case3 ces Z step: Newton residual	2.1e-15